

AI & Machine Learning — Foundations

My notes from completing an Artificial Intelligence course

1. What is Artificial Intelligence (AI)?

Artificial Intelligence is the broad field of building machines and software that can perform tasks which normally require human intelligence — such as recognising patterns, making decisions, and understanding language. Most modern AI systems are powered by **Machine Learning**.

2. What is Machine Learning (ML)?

Machine Learning is the process of teaching a computer to learn patterns from **data** — rather than being explicitly programmed with fixed rules — and to make decisions or predictions based on those patterns. Its main subsets are **Deep Learning** and **Generative AI**.

3. What is Deep Learning (DL)?

Deep Learning is a subset of Machine Learning that works using **artificial neural networks**, structures loosely inspired by how neurons in the human brain are connected. It excels at learning from large volumes of complex data such as images, audio, and text.

4. How Machine Learning Works: Training vs. Inference

A machine learning workflow generally has two stages:

- **Training:** the model learns patterns from historical, labeled data (e.g. past loan applications marked Accept/Reject).
- **Inference (prediction):** the trained model is applied to a new, unseen case to predict an outcome.

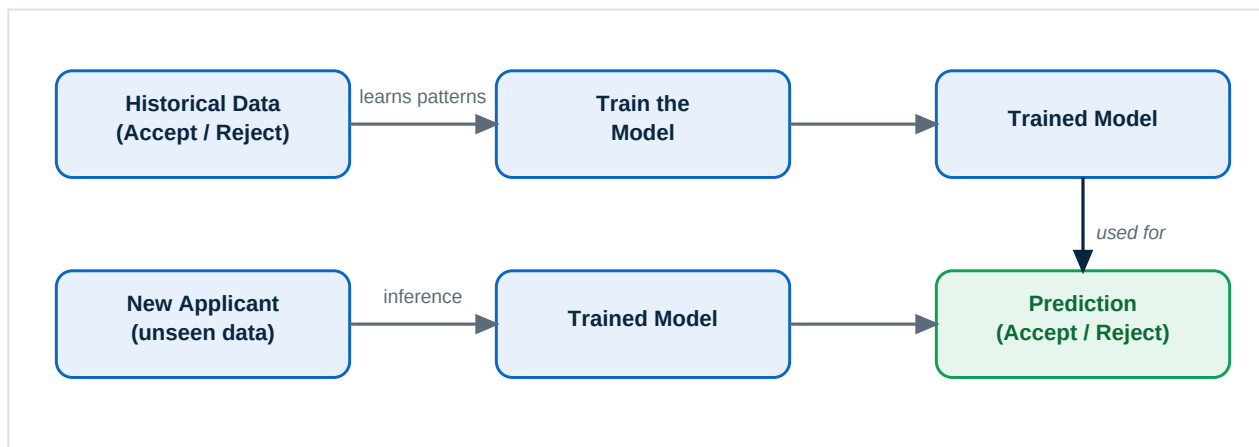


Fig 1 — Training on historical data, then using the trained model for inference

5. Types of Machine Learning

Machine learning is commonly divided into three types: **Supervised**, **Unsupervised**, and **Reinforcement Learning**.

5.1 Supervised Learning

The model learns from **labeled data** — data where the correct answer is already known — and then predicts outcomes for new data.

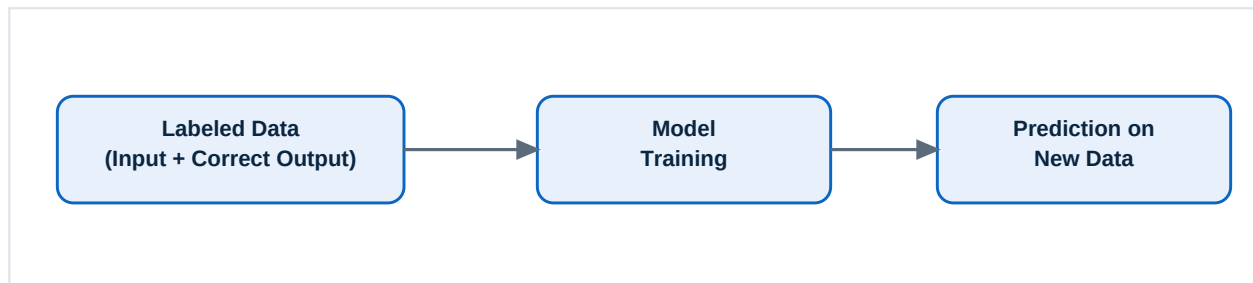


Fig 2 — Supervised learning: label data in, prediction out

Supervised learning splits into two problem types:

- **Classification** — predicting a category. Includes *binary* classification (2 classes, e.g. spam/not spam) and *multiclass* classification (3+ classes). Common algorithms: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, XGBoost.
- **Regression** — predicting a continuous number (e.g. price, salary). Common algorithms: Linear Regression, Multiple/Multivariate Regression.

5.2 Unsupervised Learning

The model learns from **unlabeled data** — there is no known correct answer — and instead finds hidden structure or groupings on its own.

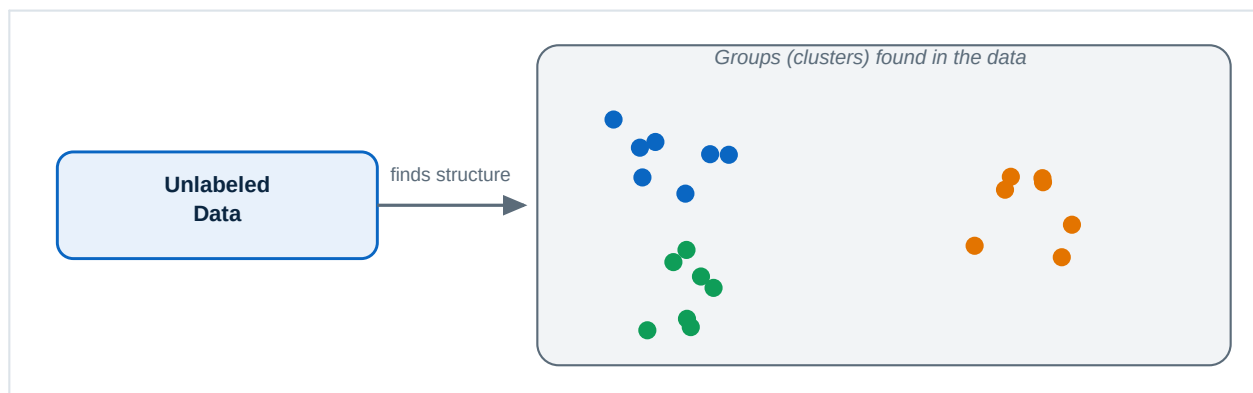


Fig 3 — Unsupervised learning: the model groups similar data points together

- **Clustering** — grouping similar data points together.
- **Association** — finding relationships between items, e.g. *Market Basket Analysis* (customers who buy X also buy Y).
- Common algorithms: Hierarchical Clustering, Principal Component Analysis (PCA), DBSCAN.

5.3 Reinforcement Learning (RL)

An **agent** learns to make decisions by interacting with an **environment**, receiving rewards or penalties based on its actions, and improving its strategy over time.

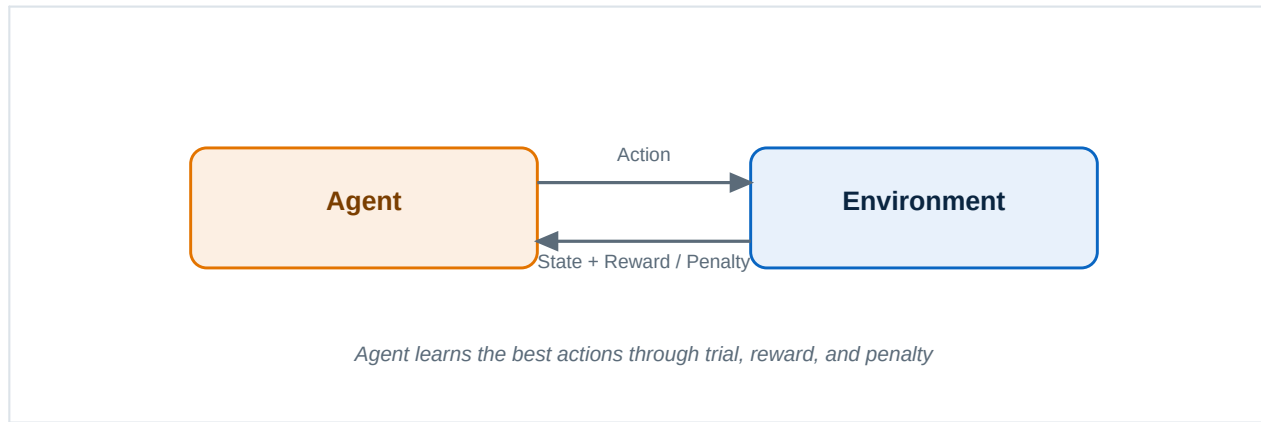


Fig 4 — The agent–environment feedback loop in reinforcement learning

Common algorithms: Q-Learning, Deep Q-Network (DQN), Policy Gradient methods, Proximal Policy Optimization (PPO).

6. Deep Learning Tools & Frameworks

- **TensorFlow** — open-source deep learning framework developed by Google.
- **PyTorch** — open-source deep learning framework developed by Meta, popular in research.
- **Keras** — a high-level API that makes building neural networks simpler (runs on top of TensorFlow).

7. Neural Network Architectures

Different problems call for different neural network designs. The four foundational architectures are:

7.1 Feedforward Neural Network (FNN)

Data moves in a single direction — from the input layer, through hidden layers, to the output layer. Simple and effective, but not well suited to sequential or time-dependent data.

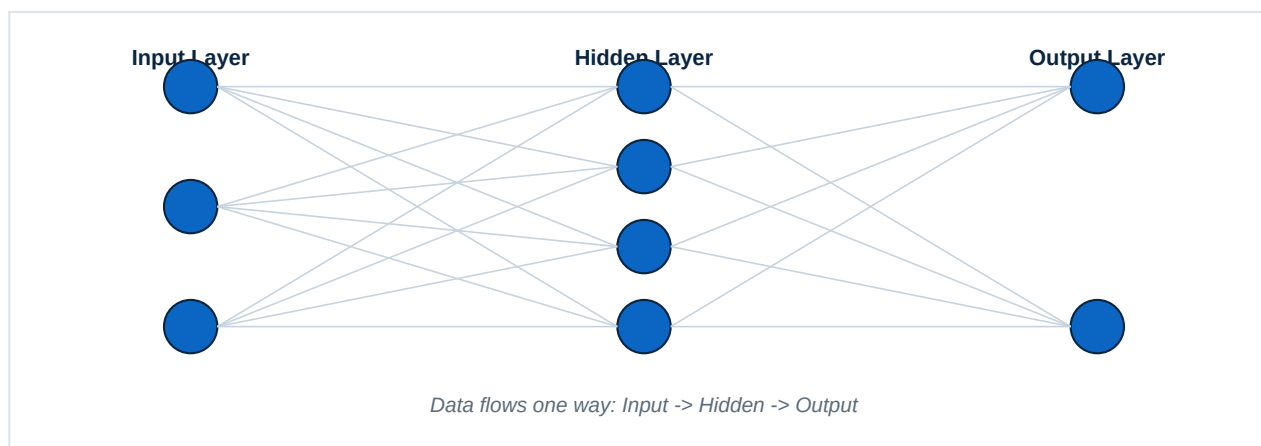


Fig 5 — Feedforward Neural Network: one-directional data flow

7.2 Recurrent Neural Network (RNN)

Has input, hidden, and output layers like an FNN, but the hidden layer includes a form of **memory** — it remembers information from previous steps and uses it to influence the current step. Good for sequences,

though it struggles to retain very long-term dependencies.

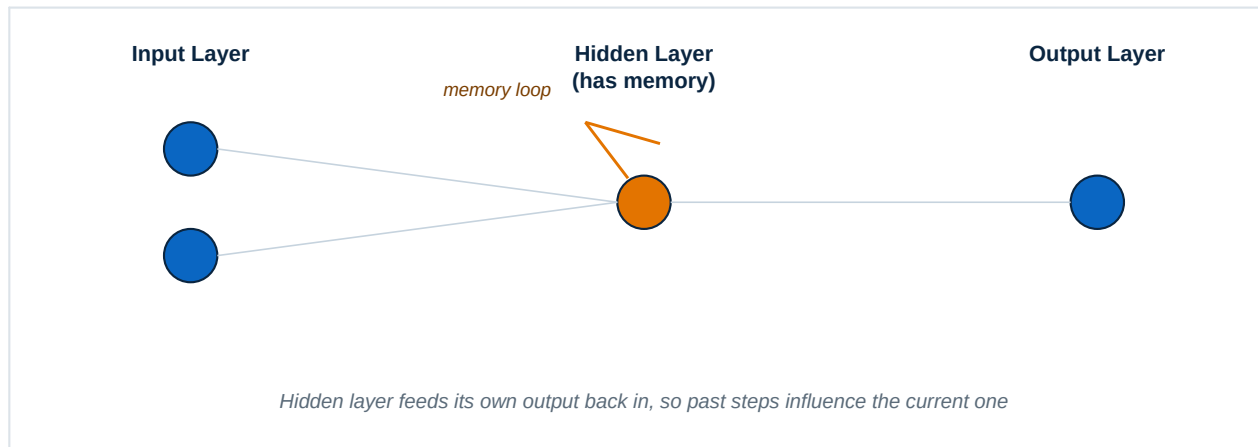


Fig 6 — Recurrent Neural Network: memory loop in the hidden layer

7.3 Convolutional Neural Network (CNN)

Designed for image data. A small **filter (kernel)** slides across the pixel grid, detecting local patterns such as edges and shapes; these are combined in deeper layers to recognise full objects.

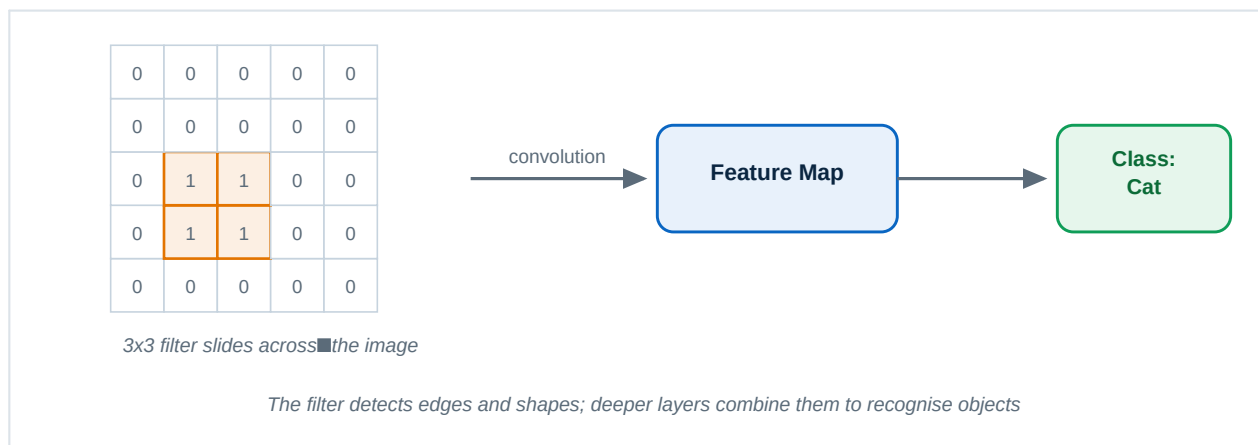


Fig 7 — CNN: a filter scans the image to build a feature map

7.4 Transformers

The architecture behind modern large language models. Uses a **self-attention** mechanism so the model can weigh how every word in a sequence relates to every other word, processing the whole sequence in parallel rather than step by step.

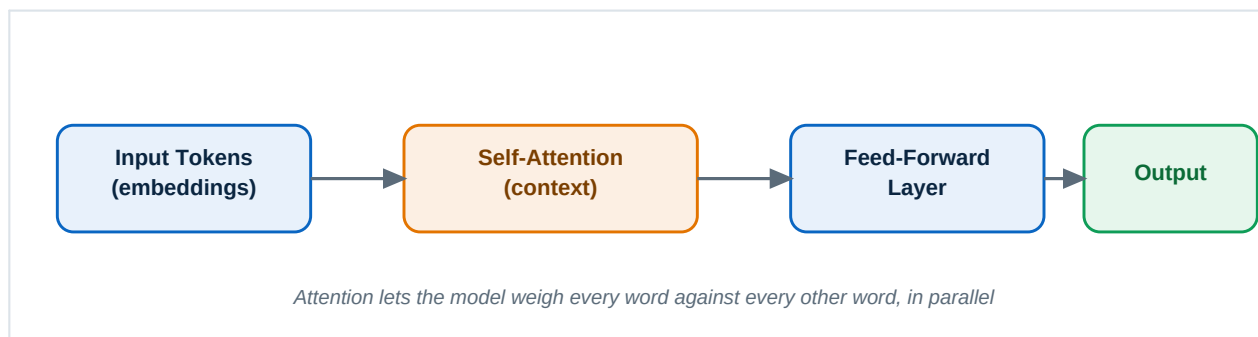


Fig 8 — Transformer: attention followed by a feed-forward layer

8. Generative AI, NLP & LLMs

- **Generative AI** — AI systems that create new content (text, images, audio, code) rather than just classifying or predicting on existing data.
- **GPT (Generative Pre-trained Transformer)** — a Transformer-based model, pre-trained on huge amounts of text, that powers many generative AI applications such as chatbots.
- **NLP (Natural Language Processing)** — the field focused on enabling machines to understand, interpret, and generate human language.
- **LLM (Large Language Model)** — a language model (often Transformer/GPT-based) trained on massive text datasets, capable of understanding and generating human-like text.

Personal notes from an AI fundamentals course — shared for anyone starting their own AI/ML learning journey.